Unit 6: Impairments, Error handling and Compression Techniques

1. What is attenuation, distortion and noise? Explain the different types of noise in communication systems. [7] 2011 [S]

2. Generate CRC Code for the data word 110101011 using the divisor 10101. The code word received as 1100100101011. Check if there are errors in the code word. [5] 2011 [F]

3. Explain how the following transmission impairments affect analog and digital data:

   i) Delay distortion

   ii) Attenuation [5] 2011 [F]

4. Define noise, delay distortion, interference and cross talk. [5] 2012 [S]

5. What do you mean by data compression technique? Explain the different error detecting techniques. [8] 2012 [S]

6. What is attenuation, distortion and noise? Explain the different types of noise in communication systems. [7] 2012 [F]

7. State various data compression techniques used in data communication. Explain one of them in detail. [7] 2013 [S]

8. What is error in data transmission? Explain the operation of checksum for error detection purpose with an examples. [7] 2013 [F]

9. List Various types of data compression techniques with their brief description. [5] 2013 [F]

10. What is redundancy? Explain the operation of CRC for error detection purpose with an example. [7] 2014 [F]

11. Explain different type of signal impairments in communication system. [8] 2014 [S], 2019 [F], 2023 [F] (New)

*Define signal Impairment

12. What is error control? Explain how hamming code for error correction works with example. [7] 2013 [S]

13. Write short notes on:

- Data compression. [5] 2015 [S], 2016 [F], 2018 [S]

- Lossless & lossy compression. [5] 2017 [F]
- Noise and its types [5] 2023 [F],2024 [F]

14. How are burst errors difficult to handle than single bit errors? Explain hamming code for error correction. [7] 2015 [F]

15. Explain CRC method for detection and correction of error. Explain how data link handles lost and duplicate frames that carry data acknowledgements. [7] 2016 [F]

16. Define errors with its types and explain the error detection technique. [8] 2016 [S], 2018 [F]

17. How is burst errors difficult to handle than single bit error? Explain why CRC is better than parity method of error detection. [7] 2017 [F], 2022 [F], 2023 [F] (New)

18. Explain attenuation and distortion with examples. If a signal at the beginning of a cable with 0.3 dB/km has a power of 2 mW, what is the power of a signal at 5 km? Comment on the result. [8] 2018 [S]

19. What is the purpose of hamming code? Consider the following message M=1010001101. The cyclic redundancy check (CRC) for this message using the divisor polynomial $\mathbf{x^5 + x^4 + x^2 + 1}$. Calculate error free message at receiver. [8] 2019 [S]

20. Explain how redundancy is introduced in parity method of error detection and correction. [7] 2020 [F]

21. Explain the need for data compression in data communication. [4] 2020 [F]

22. How does data link error control differ from flow control? Explain. [4] 2020 [F]

23. A bit stream 11010101 is transmitted using CRC method. The generator polynomial is $\mathbf{x^3 + 1}$. Show the actual bit transmitted. Suppose third bit from the left is inserted during transmission. Show that this error is detected at the receiver end. [10] 2020 [S]

24. Explain different methods for detection and correction of single bit error and burst error. [7] 2021 [F]

25. Explain different types of data compression techniques in data communication. Highlight its importance in communication. [8] 2021 [S]

26. A bit stream is 10011101 is transmitted using the standard CRC method. The generator polynomial is $\mathbf{x^3 + 1}$. Derive the actual bit string transmitted. Show that the error is detected if any one bit in the received bit stream if inverted. [7] 2023 [S]

27. Compare and contrast between lossless and lossy compression. [8] 2023 [F] (New)

28. A bit stream 1101011011 should be transmitted using the CRC method. The generator polynomial is $x^4 + x + 1$. Determine the transmitted codeword. [8] 2024 [S],2024 [F]

29. Briefly explain attenuation, distortion and noise that occurs in data communication. Why are they called transmission impairments? [7] 2024 [S] (New)

30. If the frame is $x^7 + x^5 + 1$ by the generator polynomial $x^3 + 1$, what would be the transmitted frame using CRC? [8] 2024 [S] (New)

31. Encode the given five source symbols (message) using Huffman codes and find its efficiency. [8] 2024 [S] (New)

Message	m_1	m_2	m_3	m_4	m_5
Probability	0.4	0.2	0.2	0.1	0.1

32. If the 7-bit Hamming codeword received by receiver is 1011011.Assuming the even parity, state whether the received codeword is correct or wrong. If wrong, decode the correct data word and locate the bit in error. [8] 2024 [F] (New)

33.Compress “Engineering” using Huffman codes and find its efficiency and redundancy. [8] 2024 [F] (New)

34. What is attenuation? If the frame is $x^7 + x^5 + 1$ by the generator polynomial $x^3 + 1$, what would be the transmitted frame using CRC? [8] 2025 [S], same as question 30 2024 [S]

35. A bit stream is 11010101 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$.Show the actual bit transmitted. Suppose, the third bit from the left is inverted during transmission. Show that this error is detected at the receiver’s end. [7] 2025 [S] (New)